

Implementation of a Data-Driven Model Predictive Controller for a Self-Stabilizing Bicycle

Research Project Plan

Riding a bicycle is a common activity for humans, but designing a controller to autonomously stabilize a bicycle presents significant engineering challenges. Traditional model-based controllers, such as linear quadratic regulators (LQR) and model predictive control (MPC), require an accurate system model. However, some parameters of a bicycle, i.e., the center of gravity, can be difficult to determine and may fluctuate due to varying loads or additional passengers. In this project, we aim to design and implement a data-driven MPC controller to stabilize an autonomous mini bike. The method does not identify the system model of the bike and directly uses the collected data to design the controller. The goal is to stabilize the bike at a constant speed and the bike does not fall down. Besides, we aim to keep both the lean angle and steering angle within defined safe limits for safety.

Tasks:

Introduction and background research

1. Literature Review:
 - a. Literature of Data-driven Model Predictive Control (MPC) [1, 2]
 - b. Literature about the bicycle: Modelling and Control [3, 4, 5]
 - i. Modelling of the system,
 - ii. Model-based controllers for the bike.
2. Implement Simulation for Data-driven MPC
 - a. Generate data using the model of the bike
 - b. Implement data-driven MPC with this data
3. Get familiar with the mini-bike
 - a. Read its documentation
 - b. Familiarise with the STM32 codes

Implementation and Experimenting (front-steered bicycle)

1. Run the bicycle and collect data:
 - a. Open-loop data
 - b. Closed-loop data
2. Implement data-driven State-feedback Controller
 - a. Write the STM32 codes for a state-feedback controller
 - b. Run the bicycle and collect data for the results
3. Implement DDMPC on the bike
 - a. Check appropriate solvers for the Semi-Definite Program (SDP) with C++
 - b. Integrate the solver for our problem on the STM32 microcontroller
 - c. Run the bicycle and collect data for the results
4. Evaluate the results

Optional: if there is time, implement and experiment on the **Reer-steered mode** of the bike.

Finalise

1. Write the *report*
2. Prepare and do the *presentation*

References

1. Xie, Yifan, Julian Berberich, and Frank Allgöwer. "Data-driven min-max MPC for linear systems." *2024 American Control Conference (ACC)*. IEEE, 2024.
2. Xie, Yifan, Julian Berberich, and Frank Allgöwer. "Data-Driven Min-Max MPC for Linear Systems: Robustness and Adaptation." arXiv preprint arXiv:2404.19096 (2024).
3. Cerone, Vito, et al. "Stabilization of a riderless bicycle a linear-parameter-varying approach." *IEEE control systems* 30.5 (2010): 23-32.
4. Persson, Niklas, et al. "A comparative analysis and design of controllers for autonomous bicycles." *2021 European Control Conference (ECC)*. IEEE, 2021.
5. Tapken, C., et al. "Modelling and Controlling the Kinetic and Dynamic of a Bicycle." (2018).

Timeline Plan

Task	Nov		Dec	Jan	Feb	Mar	Apr
Literature Review		2 W.					
DDMPC Simulation & Familiarise with bike			2 W.				
Run and collect data			6 W.				
Implement data-driven State-feedback				4 W.			
Implement DDMPC on the bike					6 W.		
Evaluate the results						2 W.	
Write report and presentation						6 W.	